MATHEMATICS

Analysis and approaches

Higher Level | Set B | Paper 1

Mock examination · 2025 series

Time allowed: 2 hours Maximum mark: 110

Candidate name: _____

Candidate number: _____

Date: _____

Instructions

- A calculator is not permitted.
- This paper has Section A and Section B. Both sections are compulsory.
- Answer every question. Show all working and give sufficient mathematical reasoning.
- A clean Mathematics: analysis and approaches formula booklet may be used.
- Give exact answers where appropriate. Otherwise give numerical answers to three significant figures unless instructed differently.
- Diagrams are not necessarily drawn to scale. Use the stated measurements and conditions.
- The number in square brackets shows the maximum mark for each part.

SECTION A

Question 1 [Maximum mark: 7]

Points $A(1, 1, 1)$, $B(2, -2, 5)$ and $C(3, 0, 3)$ define a triangle in three-dimensional space.

(a) Find $\overrightarrow{AB} \times \overrightarrow{AC}$. [3]

(b) Find the exact area of triangle ABC. [2]

(c) Find a Cartesian equation of the plane through A, B and C. [2]

SECTION A

Question 2 [Maximum mark: 6]

Let $f(x) = x^2 + ax + b + 2$ and $g(x) = ax^2 + bx + c - 1$, where $a \neq 0$. It is given that $f(1) = g(1)$.

(a) Find c. [2]

(b) The equations $f(x) = b + 4$ and $g(x) = -1$ have the same two roots. Find a and b. [4]

SECTION A

Question 3 [Maximum mark: 8]

In this question $\operatorname{cis}\theta = \cos\theta + i\sin\theta$.

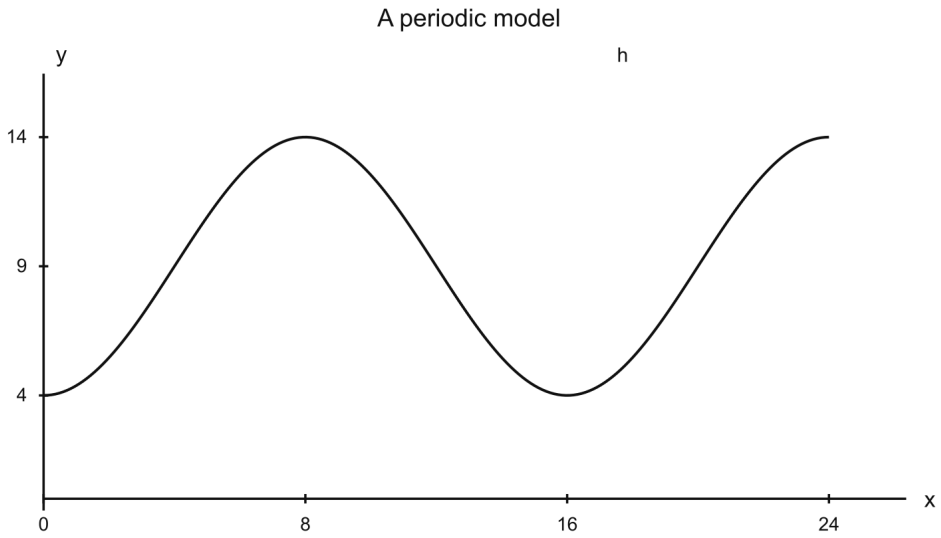
(a) Find all three solutions of $z^3 = 64i$, expressing each in the form $a + bi$. [6]

(b) Hence solve $(z + 2i)^3 = 64i$. [2]

SECTION A

Question 4 [Maximum mark: 7]

The graph shows a periodic function $h(x) = p\cos(qx) + r$, where $q > 0$. The curve starts at a minimum when $x = 0$ and completes one and a half periods between $x = 0$ and $x = 24$.



(a) Find the period. [2]

(b) Find p, q and r. [5]

SECTION A

Question 5 [Maximum mark: 6]

Solve $4^{x+1}\,3^{2x+1} = 18^{x+1}$, giving your answer in the form $a + \log_2 b$, where a and b are integers.

Show your algebraic working. [6]

SECTION A

Question 6 [Maximum mark: 7]

The function $y(x)$ satisfies $\frac{dy}{dx} + \frac{4x^3}{1+x^4}y = 5$ and $y(1) = 1$.

Find y in terms of x. [7]

SECTION A

Question 7 [Maximum mark: 7]

Let $f(x) = \frac{3}{2}x^4 - \frac{1}{3}x^3 - \frac{5}{2}x^2 + 1$.

(a) Find the gradient of the tangent to the graph of f at $x = -1$. [2]

(b) Find the other two values of x at which the tangent is parallel to the tangent in part(a). [5]

SECTION A

Question 8 [Maximum mark: 7]

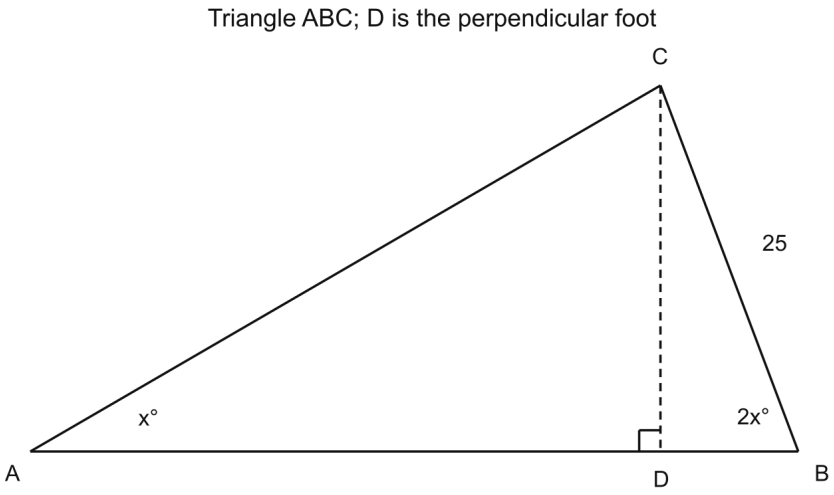
An arrangement uses each of the nine distinct digits 1 to 9 and each of three distinct symbols (dollar sign, at sign and hash sign) exactly once. Let A be the number of arrangements in which the three symbols are consecutive. Let B be the number in which no two symbols are consecutive.

By finding expressions for A and B, show that $B = 12A$. [7]

SECTION B

Question 9 [Maximum mark: 18]

In triangle ABC, $\angle CAB = x°$, $\angle ABC = 2x°$ and $BC = 25$. Point D is the foot of the perpendicular from C to AB. It is given that $\sin x = 3/5$ and $0 < x < 90$. The diagram is schematic.



(a) Find the exact values of $\cos x$ and $\sin 2x$. [4]

(b) Find $\cos 2x$ exactly. [3]

(c) Find AC. [3]

(d) Find AD. [3]

(e) Find BD. [2]

(f) Hence find AB and the area of triangle ABC. [3]

SECTION B

Question 10 [Maximum mark: 17]

The function $f : \mathbb{R} \to \mathbb{R}$ is defined by $f(x) = e^{2x} + e^x + x - 6$.

(a)(i)) Show that f is strictly increasing on its domain. [2]

(a)(ii)) Show that its graph is concave upwards everywhere. [2]

(b) On the same axes sketch $y = f(x)$, $y = f^{-1}(x)$ and $y = x$. Label the y-intercept of f and the corresponding intercept of its inverse. Show qualitatively where all three meet. [4]

(c) Find the exact coordinates of the common point of the three graphs, justifying why an intersection of f and its inverse must lie on $y = x$. [5]

(d) The graph of $y = 2f(x) + k$ passes through the origin. Find k. [2]

(e) For $x > 0$, simplify $f(\ln x) - \ln x$. [2]

SECTION B

Question 11 [Maximum mark: 20]

The sequence (u_n) is defined by $u_0 = 1$ and $u_{n+1} = 4^{n+1} - (n+1)u_n$ for $n = 0, 1, 2, \ldots$. Also define $I_n = \int_0^1 x^n e^{4x}\, dx$.

(a) Find u_1 and u_2. [2]

(b) Find I_0 exactly. [1]

(c) Use integration by parts to show that $I_{n+1} = e^4/4 - (n+1)I_n/4$. [4]

(d)(i)) Find I_1 exactly. [2]

(d)(ii)) Find I_2 exactly. [2]

(e) Hence evaluate $\int_0^1 (64x^2 + 16x + 4)e^{4x}\, dx$. [3]

(f) Prove by mathematical induction that $I_n = \dfrac{u_n e^4 + (-1)^{n+1} n!}{4^{n+1}}$ for every integer $n \geq 0$. [6]

End of question paper

MATHEMATICS

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Worked solution book

This section contains the worked solutions and marking points. Use it after completing the question paper. Award method, accuracy and reasoning marks separately, and accept mathematically equivalent valid methods.

Mark your written working against the solution book before entering self-assessed marks in the portal. A correct final answer does not always earn every method or proof mark.

Section A

1. [Maximum mark: 7]

(a) $\overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} 1 \\ -3 \\ 4 \end{pmatrix} \times \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} -2 \\ 6 \\ 5 \end{pmatrix}$. **A1 A1 A1**

[3 marks]

(b) $Area = \frac{1}{2}\left|\overrightarrow{AB} \times \overrightarrow{AC}\right| = \frac{1}{2}\sqrt{4+36+25} = \frac{\sqrt{65}}{2}$ **M1 A1**

[2 marks]

c) $-2x + 6y + 5z = d$ **M1**

$d = -2 + 6 + 5 = 9$

Hence,

$-2x + 6y + 5z = 9$ **A1**

[2 marks]

2. [Maximum mark: 6]

(a) $f(1) = g(1) \Rightarrow 1 + a + b + 2 = a + b - 1 + c \Rightarrow c = 4$. **M1 A1**

[2 marks]

(b) Equal sums: $-\frac{a}{1} = -\frac{b}{a} \Rightarrow b = a^2$ **A1**

Equal products: $-\frac{2}{1} = \frac{c}{a} \Rightarrow -2a = c \Rightarrow -2a = 4 \Rightarrow a = -2$ **M1 A1**

Hence, $b = 4$, **A1**

[4 marks]

3. [Maximum mark: 8]

(a) Let $z = r\text{cis}\theta$.

$\left(r\text{cis}\theta\right)^3 = 64\text{cis}\frac{\pi}{2} \Rightarrow r^3\text{cis}3\theta = 64\text{cis}\frac{\pi}{2}$ **M1**

$r^3 = 64 \Rightarrow r = 4$ **A1**

$3\theta = \frac{\pi}{2} + 2k\pi \Rightarrow \theta = \frac{\pi + 4k\pi}{6}$ **M1**

The solutions are $z_k = 4\text{cis}\left(\frac{\pi + 4k\pi}{6}\right),\ k = 0,1,2$ **A1**

$z_0 = 4\text{cis}\frac{\pi}{6} = 4\left(\frac{\sqrt{3}}{2} + \frac{1}{2}\text{i}\right) = 2\sqrt{3} + 2\text{i}$ **A1**

$z_1 = 4\text{cis}\frac{5\pi}{6} = 4\left(-\frac{\sqrt{3}}{2} + \frac{1}{2}\text{i}\right) = -2\sqrt{3} + 2\text{i}$

$z_2 = 4\text{cis}\frac{9\pi}{6} = 4\text{cis}\frac{3\pi}{2} = -4\text{i}$ **A1**

[6 marks]

(b) $z + 2\text{i} = z_k \Rightarrow z = z_k - 2\text{i}$ **M1**

Hence the solutions are

$z = 2\sqrt{3},\ z = -2\sqrt{3},\ z = -6\text{i}$ **A1**

[2 marks]

4. [Maximum mark: 7]

(a) From 0 to 24: 1.5 periods **M1**

Hence, Period = 16 **A1**

[2 marks]

(b) central: 9 $r = 9$ **A1**

Amplitude: 5 $p = -5$ (starts from min) **M1 A1**

Period: 16 $q = \frac{2\pi}{period} = \frac{2\pi}{16} = \frac{\pi}{8}$ **M1 A1**

[5 marks]

5. [Maximum mark: 6]

$4^{x+1}\times 3^{2x+1}=18^{x+1}$.

$4^{x}\times 4\times 3^{2x}\times 3=18^{x}\times 18$. **M1**

$\frac{4^{x}\times 3^{2x}}{18^{x}}=\frac{18}{4\times 3}$ **M1**

$\left(\frac{4\times 9}{18}\right)^{x}=\frac{3}{2}$ **A1**

$2^{x}=\frac{3}{2}$ **A1**

$x=\log_2\frac{3}{2}=\log_2 3-\log_2 2=-1+\log_2 3$. **M1 A1**

[6 marks]

Note: There are several alternative methods, but this is a quick one!

6. [Maximum mark: 7]

$\frac{dy}{dx}+\frac{4x^3}{x^4+1}y=5$, Linear with $P(x)=\frac{4x^3}{x^4+1}$, $Q(x)=5$. **M1**

Integrating factor: $I=e^{\int\frac{4x^3}{x^4+1}dx}=e^{\ln\left(x^4+1\right)}=x^4+1$ **A1**

If we multiply the differential equation by I we obtain

$Iy=\int IQdx$ **M1**

$\Rightarrow\left(x^4+1\right)y=\int 5\left(x^4+1\right)dx=\int\left(5x^4+5\right)dx=x^5+5x+c$ **A1 A1**

For $x=1, y=1$, $2=6+c\Rightarrow c=-4$ **A1**

Hence, $\left(x^4+1\right)y=x^5+5x-4$

$\Rightarrow y=\frac{x^5+5x-4}{x^4+1}$. **A1**

[7 marks]

7. [Maximum mark: 7]

(a) $f'(x) = 6x^3 - x^2 - 5x$

$f'(-1) = -6 - 1 + 5 = -2$ **M1 A1**

[2 marks]

(b) $f'(x) = -2 \Leftrightarrow 6x^3 - x^2 - 5x = -2 \Leftrightarrow 6x^3 - x^2 - 5x + 2 = 0$ **M1**

But we already know a solution $x = -1$

Using long division by $(x+1)$ **M1**

we obtain the quadratic $(6x^2 - 7x + 2)$ **A1**

Therefore, $x = \frac{2}{3}$, $x = \frac{1}{2}$ **A1 A1**

[5 marks]

8. [Maximum mark: 7]

For A,

we have 9 digits and the "group digit" (\$, @, #), that is 10 objects **M1**

Hence $A = 10! \times 3!$ **A1 A1**

For B,

The 9 digits determine 10 possible positions for the symbols \$, @, #

Hence, for the 3 symbols: $10 \times 9 \times 8$ **M1**

Then for the 9 digits: $9!$ **A1**

Therefore, $B = 10 \times 9 \times 8 \times 9!$ **A1**

$$\frac{B}{A} = \frac{10 \times 9 \times 8 \times 9!}{10! \times 3!} = \frac{9 \times 8}{3!} = \frac{9 \times 8}{3 \times 2 \times 1} = 3 \times 4 = 12$$ **A1**

[7 marks]

Section B

9. [Maximum mark: 18]

$\sin x = \frac{3}{5},\ 0° < x < 90°$

(a) By using the Pythagorean identity **OR** the triangle method **M1**

$\cos x = \frac{4}{5}$ **A2**

$\sin 2x = 2\sin x \cos x = 2 \times \frac{3}{5} \times \frac{4}{5} = \frac{24}{25}.$ **A1**

[4 marks]

(b) $\cos 2x = 1 - 2\sin^2 x = 1 - 2\left(\frac{3}{5}\right)^2 = 1 - \frac{18}{25} = \frac{7}{25}.$ **M1 A1 A1**

[3 marks]

(c) $\frac{25}{\sin x} = \frac{AC}{\sin 2x}$ **M1**

$\Leftrightarrow AC = \frac{25}{\sin x} \times \sin 2x \Leftrightarrow AC = \frac{25}{\frac{3}{5}} \times \frac{24}{25} = 40$ **A1 A1**

[3 marks]

(d) $\cos x = \frac{AD}{AC} \Leftrightarrow \frac{4}{5} = \frac{AD}{40} \Leftrightarrow AD = 32$ **M2 A1**

[3 marks]

(e) $\cos 2x = \frac{BD}{BC} \Leftrightarrow \frac{7}{25} = \frac{BD}{25} \Leftrightarrow BD = 7$ **M1 A1**

[2 marks]

(f) $AB = AD + BD = 32 + 7 = 39$ **A1**

$\text{Area} = \frac{1}{2} \times 40 \times 39 \times \sin x = 20 \times 39 \times \frac{3}{5} = 468$ **M1 A1**

[3 marks]

10. [Maximum mark: 17]

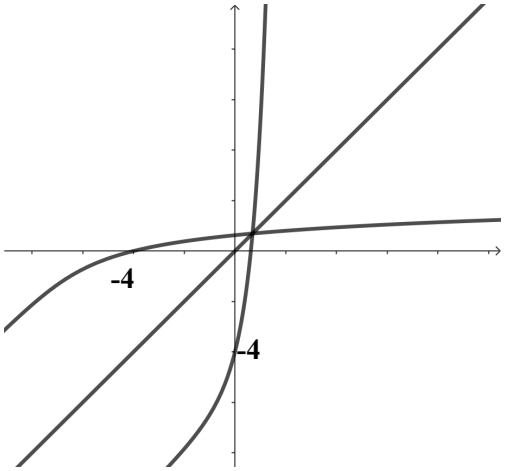
$f(x) = e^{2x} + e^x + x - 6$.

(a) (i) $f'(x) = 2e^{2x} + e^x + 1 > 0$ for all x, hence increasing. **M1 A1**

(ii) $f''(x) = 4e^{2x} + e^x > 0$ for all x, hence concave up. **A1 A1**

[4 marks]

(b)



Inverse, $y = x$ **A1 A1**

Meeting all 3 at the same point **A1**

x-intercept at -4 **A1**

[4 marks]

(c) $f(x) = f^{-1}(x) = x \Leftrightarrow f(x) = x \Leftrightarrow e^{2x} + e^x + x - 6 = x$ **M1**

$\Leftrightarrow e^{2x} + e^x - 6 = 0$ (Hidden quadratic by letting $u = e^x$) **M1**

Solutions: $e^x = 2$, $e^x = -3$ -3 (rejected) **A1 A1**

$e^x = 2 \Leftrightarrow x = \ln 2$ **A1**

[5 marks]

(d) Using transformations: $(0,-4) \to (0,-8) \to (0,-8+k)$, $k = 8$ **M1 A1**

OR

$y = 2\left(e^{2x} + e^x + x - 6\right) + k$, so for $x = 0, y = 0$, $k = 8$ **M1 A1**

[2 marks]

(e) $f(\ln x) - \ln x \quad = e^{2\ln x} + e^{\ln x} + \ln x - 6 - \ln x = x^2 + x - 6$ **M1 A1**

[2 marks]

11. [Maximum mark: 20]

$u_0 = 1$ and $u_{n+1} = 4^{n+1} - (n+1)\cdot u_n$ for $n \in \mathbf{N}$.

(a) $u_1 = 4^1 - u_0 = 4 - 1 = 3$ **A1**

$u_2 = 4^2 - 2u_1 = 16 - 6 = 10$. **A1**

[2 marks]

$$I_n = \int_0^1 x^n \, \mathrm{e}^{4x} \, \mathrm{d}x$$

(b) $I_0 = \int_0^1 \mathrm{e}^{4x} \, \mathrm{d}x = \left[\frac{\mathrm{e}^{4x}}{4}\right]_0^1 = \frac{\mathrm{e}^4 - 1}{4}$. **A1**

[1 mark]

(c) $I_{n+1} = \int_0^1 x^{n+1} \, \mathrm{e}^{4x} \, \mathrm{d}x = \left[x^{n+1} \frac{\mathrm{e}^{4x}}{4}\right]_0^1 - \int_0^1 (n+1)x^n \frac{\mathrm{e}^{4x}}{4} \mathrm{d}x$ **M1 A1 A1**

$\left[\frac{\mathrm{e}^4}{4} - 0\right]_0^1 - \frac{n+1}{4}\int_0^1 x^n \, \mathrm{e}^{4x} \mathrm{d}x$ **A1**

$\frac{\mathrm{e}^4}{4} - \frac{n+1}{4} I_n$.

[4 marks]

(d) (i) $I_1 = \frac{\mathrm{e}^4}{4} - \frac{1}{4} I_0 = \frac{\mathrm{e}^4}{4} - \frac{1}{4} \times \frac{\mathrm{e}^4 - 1}{4} = \frac{4\mathrm{e}^4 - \mathrm{e}^4 + 1}{16} = \frac{3\mathrm{e}^4 + 1}{16}$ **M1 A1**

(ii) $I_2 = \frac{\mathrm{e}^4}{4} - \frac{2}{4} I_1 = \frac{\mathrm{e}^4}{4} - \frac{2}{4} \times \frac{3\mathrm{e}^4 + 1}{16} = \frac{16\mathrm{e}^4 - 6\mathrm{e}^4 - 2}{64}$. **M1 A1**

$= \frac{10\mathrm{e}^4 - 2}{64} \quad \left(= \frac{5\mathrm{e}^4 - 1}{32}\right)$

[4 marks]

(e) $\int_0^1 (64x^2 + 16x + 4)\mathrm{e}^{4x} \, \mathrm{d}x = 64I_2 + 16I_1 + 4I_0 =$ **M1 A1**

$= \left(10\mathrm{e}^4 - 2\right) + \left(3\mathrm{e}^4 + 1\right) + \left(\mathrm{e}^4 - 1\right) = 14\mathrm{e}^4 - 2$ **A1**

[3 marks]

(f) $I_n = \dfrac{u_n \, e^4 + (-1)^{n+1} n!}{4^{n+1}}$, for any $n \in \mathbf{N}$.

For $n = 0$,

LHS $= I_0 = \dfrac{e^4 - 1}{4}$, RHS $= \dfrac{u_0 \, e^4 - 1}{4} = \dfrac{e^4 - 1}{4}$, true. **A1**

Assume it is true for $n = k$,

i.e. $I_k = \dfrac{u_k \, e^4 + (-1)^{k+1} k!}{4^{k+1}}$ **M1**

Prove it is true for n = k + 1,

i.e. $I_{k+1} = \dfrac{u_{k+1} \, e^4 + (-1)^{k+2} (k+1)!}{4^{k+2}}$ **A1**

Proof:

LHS = $I_{k+1} = \dfrac{e^4}{4} - \dfrac{k+1}{4} I_k = \dfrac{e^4}{4} - \dfrac{k+1}{4}\left(\dfrac{u_k \, e^4 + (-1)^{k+1} k!}{4^{k+1}} \right)$ **M1**

$= \dfrac{e^4}{4} - \left(\dfrac{(k+1) u_k \, e^4 + (-1)^{k+1} (k+1)!}{4^{k+2}} \right)$

$= \dfrac{4^{k+1} e^4 - (k+1) u_k \, e^4 - (-1)^{k+1} (k+1)!}{4^{k+2}}$

$= \dfrac{\left(4^{k+1} - (k+1) u_k\right) e^4 + (-1)^{k+2} (k+1)!}{4^{k+2}}$ **A1**

$= \dfrac{u_{k+1} \, e^4 + (-1)^{k+2} (k+1)!}{4^{k+2}}$ = RHS

The statement is true for $n = 0$, and assuming it is true for $n = k$, we proved it is also true for $n = k + 1$. Therefore, by mathematical induction it is true for any $n \in \mathbb{N}$. **R1**

[6 marks]